Heredity (Set A) — MCQ Worksheet

Science • Chapter 8 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. The father of genetics is:

(A) Charles Darwin
(B) Gregor Mendel
(C) Watson and Crick
(D) Robert Hooke

Q2. Mendel performed his experiments on:

(A) Mice
(B) Pea plants
(C) Drosophila
(D) Bacteria

Q3. A unit of inheritance is called a:

(A) Chromosome
(B) Gene
(C) Protein
(D) Cell

Q4. Alternative forms of a gene are called:

(A) Alleles
(B) Loci
(C) Phenotypes
(D) Mutations

Q5. In a monohybrid cross of TT × tt, the F1 generation is:

(A) All TT
(B) All tt
(C) All Tt
(D) TT, Tt and tt in equal numbers

Q6. In a monohybrid cross, the F2 phenotypic ratio is:

(A) 1:1
(B) 3:1
(C) 9:3:3:1
(D) 2:1

Q7. In a monohybrid cross the F2 genotypic ratio is:

(A) 1:2:1
(B) 3:1
(C) 9:3:3:1
(D) 1:1

Q8. In a dihybrid cross the F2 phenotypic ratio is:

(A) 3:1
(B) 1:2:1
(C) 9:3:3:1
(D) 1:1:1:1

Q9. The trait that appears in F1 generation is called:

(A) Recessive
(B) Dominant
(C) Co-dominant
(D) Incomplete

Q10. Mendel's Law of Segregation states that:

(A) Alleles separate during gamete formation
(B) Genes never separate
(C) Only dominant alleles pass
(D) Sex determines all traits

Q11. Number of chromosomes in a normal human cell is:

(A) 22
(B) 23 pairs (46)
(C) 24 pairs (48)
(D) 20 pairs (40)

Q12. The sex chromosomes in a human female are:

(A) XY
(B) XX
(C) XO
(D) YY

Q13. The sex of a human child is determined by:

(A) The mother
(B) The father
(C) Both equally
(D) The environment

Q14. A human male produces sperms with:

(A) Only X chromosome
(B) Only Y chromosome
(C) Either X or Y chromosome
(D) Both X and Y in same sperm

Q15. Probability of having a male child in humans is:

(A) 1/4
(B) 1/3
(C) 1/2
(D) 2/3

Q16. Acquired traits are NOT passed on to offspring because:

(A) They are too weak
(B) They do not affect DNA of germ cells
(C) Only humans pass traits
(D) They are recessive

Q17. Variations are useful because:

(A) They keep populations identical
(B) They allow species to survive in changing environments
(C) They are always harmful
(D) They cause mutations only

Q18. In Mendel's pea cross, the recessive trait reappears in:

(A) F1 generation
(B) F2 generation
(C) Both F1 and F2
(D) Never

Q19. If pure-breeding round (RR) yellow seeds are crossed with wrinkled (rr) green ones, F1 will be:

(A) All wrinkled green
(B) All round yellow
(C) 50% round, 50% wrinkled
(D) All round green

Q20. Genes are located on:

(A) Chromosomes
(B) Cell membrane
(C) Cytoplasm only
(D) Mitochondria only

Q21. Number of autosomes in a human cell is:

(A) 22 pairs
(B) 23 pairs
(C) 21 pairs
(D) 20 pairs

Q22. Variation in an organism arises because of:

(A) Imperfect DNA copying and sexual recombination
(B) Only diet
(C) Only environment
(D) No genetic factors

Q23. In sexual reproduction, the chromosomes from each parent are present in the gamete in:

(A) Half the number
(B) The full number
(C) Double the number
(D) Random number

Q24. A homozygous tall pea plant has alleles:

(A) Tt
(B) TT
(C) tt
(D) Tt or TT

Q25. In Mendel's experiments, the parental generation is called:

(A) F0
(B) F1
(C) F2
(D) P generation

— End of Worksheet —